

Two tokens need a route

Learning intention

Describe how absorbed nutrients and water reach body tissues and distinguish this from the waste route.

One token is glucose. One is water.

How can both reach a muscle tissue?

Recover the two system jobs

R1. What is digestion?

R2. What does the heart pump?

R3. Where does blood gain oxygen?

I separate breaking down from crossing

A soluble nutrient crosses the gut wall



Particles and wall are not to scale. This shows absorption, not a literal hole or a whole-food filter.

Digestion changes some food substances. Absorption moves substances across the gut wall. I follow a small glucose molecule.

I follow the transport job

Water and a soluble nutrient travel

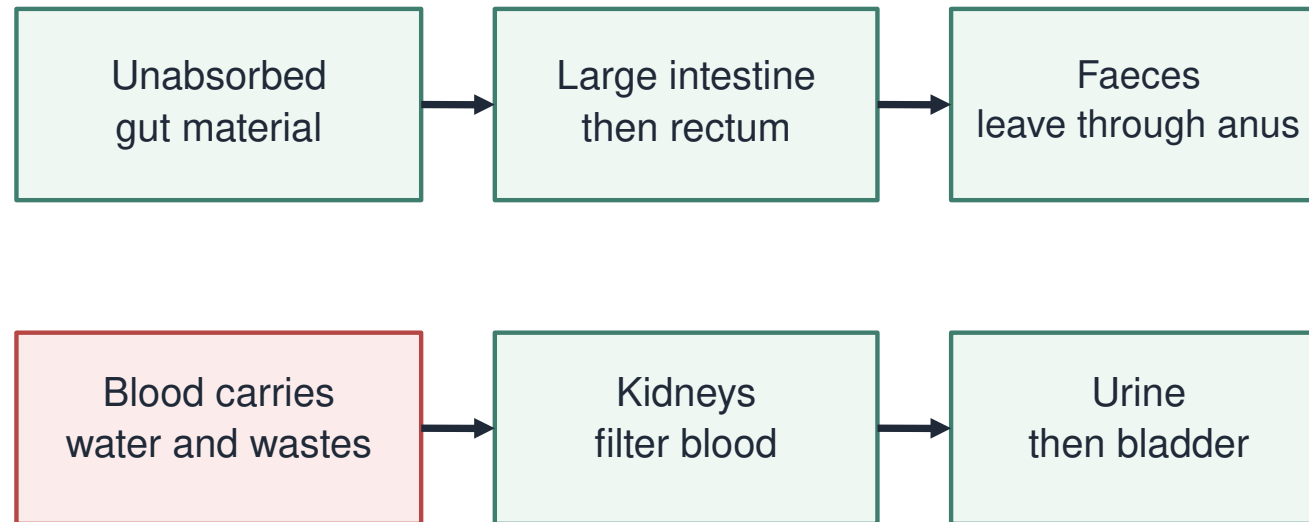


Circulation carries substances onwards. This job map omits the detailed liver, heart and lung route.

Water and many nutrients travel in blood plasma. Blood moves towards body tissues. Water can be absorbed without digestion.

I separate two waste routes

Faeces and urine take different routes



No direct intestine-to-bladder tube is shown or implied.

Unabsorbed gut material can leave as faeces. Kidneys filter blood and make urine. There is no direct gut-to-bladder tube.

Delivery desk: assign the tokens

W1. Match each token to its next process. Use cards A–F on page 1. Explain using “break down”, “cross” or “carry”.

Token position	Choose next process
Large starch molecule in gut	Digestion / circulation
Glucose at gut wall	Absorption / chewing
Dissolved glucose in blood	Transport / swallowing
Water in intestine	Absorption / digestion

The incorrect shortcut

Diagram claim: “Drink → stomach → bladder.”

W2. Replace the middle route.

Which organ makes urine from blood?

Choose the right exit route

A food particle remains unabsorbed.

W3. Does it enter blood or continue through the gut?

Explain how this differs from urine.

A water molecule is in the small intestine

Which next step can happen?

- A** It must be digested before absorption.
- B** It can be absorbed across the gut wall.
- C** It goes through a direct pipe to the bladder.

Create the delivery explanation

Choose one response route.

Explain a glucose journey and a water journey.

Keep the two waste routes separate.

Use three different verbs

Digestion: break down.

Absorption: cross a surface.

Transport: carry onwards.

Check the substance at each arrow

Is it whole food or a small molecule?

Is it inside the gut, blood or a tissue?

Have you named the process?

Trace and challenge

Trace one partner or teacher model.

Find a clear process label.

Ask for a missing step or clarification.

Choose a closer look

Would magnifying gut wall or kidney help next?

Explain which unanswered question it would address.

Three steps, three meanings

E1. What does absorption mean?

E2. What carries many dissolved nutrients?

E3. Which organs make urine from blood?